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MILITARY SPECIFICATION

PROBES, CAPACITOR TYPE, FUEL QUANTITY GAGE, GENERAL SPECIFICATION FOR

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers certain design and all performance requirements for the procurement of quick disassembly types of tank probes and compensator probes for use with capacitance type fuel quantity systems.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-S-3136	Standard Test Fluids Hydrocarbon
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-T-4687	Tester, Capacitor Type, Fuel Quantity Gage Tank Unit, Type MD-2A
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 91/96, 100/130, 115/145

FSC 6680

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MIL-J-5624	Jet Fuel, Grades JP-3, JP-4 and JP-5
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-G-7817	Gage, Fuel Quantity, Capacitor Type, Uncompensated, General Specification for
MIL-G-7818	Gage, Fuel Quantity, Capacitor Type, Compensated, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-G-7940	Gages, Fuel Quantity, Capacitor Type, Installation and Calibration of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-G-8798	Gage System, Fuel Quantity, Capacitor Type, Nonvacuum Type, Integrally Lighted, General Specification For
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-C-25516	Connectors, Electrical, Miniature, Shielded or Unshielded, Environmental Resisting Type, General Specification For
MIL-F-25656	Fuel, Aircraft Turbine and Jet Engine, Grade JP-6
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-1	Standard for Laboratory Atmospheric Conditions for Testing

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS33586	Metals, Definition of Dissimilar

Air Force - Navy Aeronautical

AN4130	Fitting, Synthetic Tank Single Plane Flush and Recessed

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2.2 Availability of Documents -

(1) When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Preproduction - The probe furnished under this specification shall be a product, samples of which have been tested and have passed the preproduction tests specified herein.

3.2 Selection of Government Documents - Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial Parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts available. In any case, commercial utility parts, such as screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard Parts - With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical Materials - Noncritical materials shall be used where practicable. Where the use of critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the probe may be subject when in storage or during normal service life.

3.3.2.1 Dissimilar Metals - Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other.

3.3.2.2 Zinc and Magnesium - Neither zinc nor magnesium metal or plating shall be used in contact with fuel.

3.3.3 Nonmagnetic Materials - Nonmagnetic materials shall be used for all parts of the probe except where magnetic materials are essential.

* 3.3.4 Fungus-Proof Material - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed case, fungicidal treatment will not be necessary.

* 3.3.5 Corrosive Fumes - Materials used in the interior construction of the instrument shall not produce corrosive or deleterious fumes.

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3.4 Design and Construction - The tank probe shall be designed to sense the quantity of usable fuel in a given aircraft fuel tank as a function of the electrical capacitance as the fuel level changes between the electrodes. The probe shall be designed for use with a null-balancing, 3-wire capacitance bridge circuit such as those specified in Specifications MIL-G-7817, MIL-G-7818 and MIL-G-8798.

3.4.1 The compensator probe shall be designed to sense the dielectric constant of the fuel for use in compensated gage systems. It shall not be designed or constructed as an integral part of the tank probe but as an entirely separate unit.

3.4.2 Accessories such as level sensing devices, if used, shall not be built as integral parts of the probes.

3.4.3 No part of the bridge circuit i.e., transformers, etc., shall be an integral part of probe.

3.4.4 The probe shall be designed so that it may be disassembled or reassembled in 1/4 manhour without the use of special tools.

3.4.5 The probes shall be so constructed that no parts will work loose in service. They shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation and service.

3.5 General - The tank and compensator probes shall be suitable for use in gaging aircraft fuels covered by Specifications MIL-G-5572, MIL-J-5624, and MIL-F-25656.

3.6 Types of Probes -

3.6.1 Tank Probe - The basic probe shall consist of a suitable capacitative arrangement of conductors, such as concentric cylinders electrically insulated from each other and arranged to form a 3-wire capacitor. The probe shall be so designed that only one shielded circuit lead is utilized in the aircraft wiring. It shall be so constructed as to permit fuel into which the probe has been immersed to rise freely to its natural level between the conductor, to insure that up to this level, the fuel will serve as the dielectric substance. Above the level of the fuel, a mixture of air and vapor will be the dielectric substance. Unless otherwise approved, nonsensing portions of the sensing element shall not exceed the following dimensions: 0.600 inch at the flange end and 0.400 inch at the opposite end for externally top-mounted and top flush mounted probes; and 0.400 inch at both ends for internally mounted probes; 0.850 inch shall be allowed for the flanged end only for bottom mounted probes, unless otherwise approved by the procuring activity. Any other length of the nonsensing portions of bottom externally mounted probes at the flange and shall be subject to the approval of the cognizant military activity. The probe shall also be constructed as to preclude the accumulation of deposits and sediments between the conductors. Openings at the bottom ends of all probes shall be as large as possible to eliminate, to the greatest degree possible, the retention of water or contaminants within the probes. The minimum allowable distance between the conductors shall be 0.125 inch. The probes shall be designed to make electrical connectors readily accessible for testing the probe, using the MD-2 or MD-2A field tester of Specification MIL-T-4687. The materials used in the construction of the probe shall be impervious to aircraft fuels, water, water soluble detergents, alcohols and chlorinated hydrocarbon solvents that might be used in cleaning.

3.6.2 Compensator Probe - The compensator probe shall meet the same general requirements as the tank probe. It shall be designed to extend a minimum distance up from the bottom of the fuel tank in which it is installed. The probe shall be designed to make electrical connections readily accessible for testing the probe, using the MD-2 or MD-2A field tester.

3.7 Probe Construction Details -

3.7.1 Mounting Types - The probe designs shall include externally mounted types, internally mounted types, and flush-mounted types, as required for specific installations. The probe shall be suitable for installation in accordance with Specification MIL-G-7940.

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3.7.2 Mounting Flange - The external top-mounted, and bottom-mounted types, unless otherwise authorized by the cognizant military activity, shall be designed and constructed for installation in fuel tank fittings which have dimensions conforming to AN4130. The flanges shall conform to size 4, 5, or 6. The means provided for mounting the other types shall be entirely suitable for the purpose and shall be subject to approval by the cognizant military activity. Flush-mounted types of probes shall be entirely suitable for the conditions of flexure, stress, and temperature encountered in the aircraft.

3.7.3 Connector Housing - For externally mounted type probes, the projection of the connector housing beyond the flange surface shall be held to a minimum practicable distance. However finger accessibility for connection and disconnection of the mating cable shall be adequate.

3.7.4 Seal - The externally and flush-mounted-type-probe head shall be sealed to prevent the leakage of fuel from the inside of the tank to the outside.

3.8 Electrical Cable - The wire and cable used in the tank probe and within the fuel cell shall be suitable for the purpose intended. For purposes of this specification, wire and cable shall be classified into the following catagories:

(1) General purpose - MIL-W-16878, Type B

(2) High temperature - MIL-W-16878, Type E

(3) Fuel resistant - MIL-W-16878, Type E

General purpose wire and cable may be used where temperatures not exceeding 90°C will be encountered and where there is no exposure to fuel. High temperature wire and cable must be used where temperatures ranging from 90°C to 200°C may be encountered. Type E wire and cable which must be used within the probes or fuel cells, shall have been tested for fuel resistance and found suitable for use in aircraft fuels.

3.8.1 Cable Length - Accuracy and sensitivity of the gage system shall not be adversely affected by the use of the probe interconnecting cables within the limits of from 5 to 200 feet.

3.8.2 Cable Assemblies - Each assembled cable and connector shall be capable of withstanding a pull of 30 pounds.

3.8.3 Electrical Connectors - The electrical connectors used on the probes shall be in accordance with Specification MIL-C-25516. The polarity of the coaxial connectors shall be as follows:

Polarity of MIL-C-25516	Application
F	High impedance
A	Low impedance-Tank probe
B	Low impedance-Compensator

3.9 Calibration Adjustment - The probes shall be so constructed that calibration adjustment shall not be required; i.e., the difference between the capacitance measured before disassembly and after reassembly of the probe shall not exceed 0.2 percent of the initial value. This is in addition to the tolerance of the master test equipment.

3.10 Performance - The probe shall perform satisfactorily when subjected to the test requirements specified in Section 4.

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3.11 Physical Requirements -

3.11.1 Finishes - Finishes and protective coatings which will crack, chip or scale during normal service or are affected by extremes of atmospheric or environmental conditions, shall not be used.

* 3.11.2 Magnesium Alloy Parts - Magnesium alloy parts shall be treated in accordance with Specification MIL-M-3171. When abrasion resistance is a factor, an anodic treatment shall be used. The treatment shall be approved by the procuring activity.

* 3.11.3 Alluminum Alloy Parts - Where practicable, alluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625 except that small holes, pipe threads and case inserts need not be anodized. Alluminum alloys which do not anodize satisfactorily shall be coated with chemical film in accordance with Specification MIL-C-5541.

3.11.4 Iron or Steel Parts - Where practicable, iron or steel parts shall be zinc or cadmium plated, in accordance with Specifications QQ-Z-325 and QQ-P-416, respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated.

3.11.5 Screw Threads - Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.11.6 Assembly Screws - Assembly screws and bolts shall be tight. The word "tight" means that the screws, bolts and/or nuts cannot be appreciably tightened further, without damage or injury to the threads.

* 3.11.7 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872.

3.11.8 Riveting - Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

* 3.11.9 Dimensions - Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the instrument, they shall be held or limited accordingly.

3.11.10 Workmanship - The probe including all parts and accessories, shall be constructed and finished to produce an instrument free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregation of coils, marking of parts and assemblies, welding and brazing, painting, plating, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges. Conductive coatings on all probe elements shall be suitable for the purpose intended and the tubes shall be suitably protected to withstand exposure to fuels, water, etc.

3.11.11 Cleaning - The instrument shall be thoroughly cleaned of loose, spattered or excess solder, resin flash that may crumble, metal chips and other foreign matter, after final assembly.

3.12 Identification of the Product -

3.12.1 Nameplate - A nameplate shall be securely attached to the exterior of the probe and shall be legibly and durably marked in accordance with MIL-STD-130. The probe dry capacitance shall be printed on the nameplate.

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*3.12.2 Manufacturer's Part Number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.12.3 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.12.4 Use of Military Designations - Military designations shall not be applied to a product, except for Qualification Test Samples, nor referred to in correspondence or sales matter, until notification has been received from the activity responsible for qualification.

*3.12.5 Installation Instructions - The contractor shall furnish with each instrument one copy of the installation instructions with illustrations and diagrams. Prior to printing, two copies shall be furnished to the Bureau of Naval Weapons for approval. Whenever possible the instructions shall be arranged to require only one sheet of paper either 8-1/2 by 11 or 11 by 17 inches.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests - The Inspection and Testing of the probes shall be classified as follows:

(a) Preproduction Tests - Preproduction tests are those tests performed after award of contract on sample instruments that are representative in design, performance and configuration of the item which will be produced on the manufacturer's production line.

(b) Acceptance Tests - Acceptance tests are those tests performed on instruments manufactured and submitted for acceptance under contract.

4.3 Preproduction Tests - The preproduction tests of the probes shall consist of all of the tests of this specification specified under the paragraph headed Test Methods.

4.3.1 Preproduction Test Samples - The preproduction test samples shall consist of one of the following:

(a) Two 24 inch length probes having a capacitance distribution as specified in Table I.

(b) Two 48 inch length probes having a capacitance distribution as specified in Table II.

(c) Two compensator probes of typical design.

The samples submitted shall not have been previously tested except for the Individual Tests. The manufacturer shall submit certified test results showing conformance with all the requirements and tests of this specification on samples other than the ones submitted for preproduction testing. The samples and certified test report shall be submitted to the testing activity designated in the contract or order.

4.3.2 Test Sample Identification - The test samples shall be plainly identified by securely attached, durable tags marked with the information listed below.

Sample for Preproduction Testing
_____ (Name of item)
Submitted by (Manufacturer's name, date) for Preproduction
Testing in accordance with Specification MIL-P-22871(WEP)
under contract or order number _____.
Type of Probe (Compensator or Tank)
Manufacturer's Part Number

*4.4 Acceptance Tests - The acceptance tests of the instruments shall consist of the Individual Tests and the Sampling Plans of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

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4.4.1 Individual Tests - The Individual tests shall consist of the following tests conducted on each probe submitted for acceptance:

Inspection
Tank Probe Dry Capacitance
Compensator Probe Dry Capacitance
Leakage
Tank Probe Calibration
Capacitance Stability
Dielectric Strength

4.4.2 Sampling Plans and Tests -

*4.4.2.1 Sampling Plan A - Samples shall consist of three probes selected at random by the inspector from each lot of 100 or less probes which have passed the Individual tests. A lot shall consist of identical probes with the same manufacturer's part number, manufacturered under substantially the same conditions and submitted at substantially the same time. Probes which have been subjected to the sampling plan A tests shall not be delivered on contract until they have been reconditioned and resubmitted to all the Individual Tests. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

_____ (Name of item)
Submitted by (manufacturer's name, date) for production acceptance
(Sampling Plan A) testing in accordance with Contract/Order No. _____

4.4.2.1.1 Sampling plan A tests - The Sampling plan A tests of the probes shall consist of the following tests conducted on each sample:

Individual Tests
Capacitance at Low Temperature
Capacitance at High Temperature
Vibration

*4.4.2.2 Sampling plan B - Unless otherwise specified, three probes selected at random by the inspector from the first 15 probes produced on contract and submitted within 10 days after manufacture shall be forwarded to the laboratory specified herein (see section 6). Probes which have been subjected to the Sampling Plan B Tests shall not be delivered on contract until they have been reconditioned and resubmitted to all the Individual Tests. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

_____ (Name of item)
Submitted by (manufacturer's name, date) for production acceptance
(Sampling Plan B) testing in accordance with Contract/Order No. _____

4.4.2.2.1 Sampling plan B tests - The sampling plan B tests of the probe shall consist of the following tests conducted on each sample:

(a) Sampling Plan A Tests

(b) Low Temperature Exposure

(c) High Temperature Exposure

(d) Water Immersion

(e) Humidity

(f) Fungus Resistance

(g) Salt Spray

(h) Fluid Resistance

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4.4.3 Rejection and Retest - The instructions on reworking, refurbishing or resubmitting rejected items shall be as specified in the General Provisions of the contract or order.

4.5 Test Conditions -

4.5.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of FED-STD-1.

4.5.2 Position - Unless otherwise specified, the probe shall be tested in its normal operating position.

4.5.3 Master Test Standard -

4.5.3.1 Capacitance Bridge - The master capacitance bridge specified herein shall be used to measure the capacitance tank probes and compensator probes. The accuracy of the bridge at an ambient temperature of 25°C shall be 0.2 percent for capacitances in excess of 50 uuf and 0.1 uuf for capacitances less than 50 uuf.

4.5.3.1.1 Tolerance - In all subsequent tests and measurements specified herein, the tolerance of the measuring equipment shall be added to the specified tolerance for the given test or measurement. The test frequency shall be 400 cps ± 2.0 percent.

4.5.3.2 Resistance Measurement - DC resistance measurements may be made with an MD-2 or MD-2A tester or with a similar device of equal or greater sensitivity.

4.6 Test Methods -

4.6.1 Inspection Test - Each probe shall be examined to determine conformance with all the requirements of this specification not covered by tests.

4.6.2 Reference Capacitance Test - The observed readings in the Tank Probe Dry Capacitance and Compensator Probe Dry Capacitance Tests will be referred to hereinafter as "reference capacitance".

4.6.3 Tank Probe Dry Capacitance Test - The dry capacitance of each tank probe shall be determined by means of the master capacitance bridge. The measured value shall not differ from the respective value specified by the manufacture by more than 1.0 percent, or 0.5 uuf, whichever is greater. This test may be combined with the Tank Probe Calibration test.

4.6.4 Compensator Probe Dry Capacitance Test - The dry capacitance of each compensator probe shall be determined by means of the master capacitance bridge. The measured value shall be determined by means of the master capacitance bridge and shall not differ from the value specified by the manufacturer by more than 1.5 percent, or 0.5 uuf, whichever is greater.

4.6.5 Probe Leakage Test - Tank probes and compensator probes of the externally mounted type shall be installed in a suitable pressure chamber equipped with the proper size flange fittings. A pressure of 30 psi shall be applied within the pressure chamber, and no leak shall be detected on the outer side of the tank probe mounting flange and connector head assembly. A suitable liquid shall be used for detecting the presence of air bubbles. The duration of this test shall be 5 minutes. A leak shall be defined as a periodic bubbling from a specific area at a frequency of at least one bubble each minute.

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4.6.6 Tank Probe Calibration Test - The dry capacitance of each of the tank probes shall be determined by means of the master capacitance bridge. Each of the tank probes shall be individually immersed in a representative fuel sample to the levels specified in Table I, II, or III, as applicable, and the added capacitance, attributable to the fuel, shall be determined. The added capacitance values, expressed as percentages of the total added capacitance for complete immersion, shall agree within 1.5 percent with the values specified in Table I, II, or III, as applicable.

TABLE I

CHARACTERIZATION OF 24 INCH LENGTH TANK PROBES FOR QUALIFICATION TESTS

Distance of fuel level below top of sensing portion (inches)	Dry capacitance per unit length (relative units)	Increase over dry capacitance (percent of total)
0	3.5	100.0
1	3.5	96.5
2	3.5	93.0
3	3.5	89.5
4	3.5	86.0
5	3.5	82.5
6	3.5	79.0
7	3.5	75.5
8	3.5-7.5	72.0
9	7.125	64.69
10	6.750	57.75
11	6.375	51.19
12	6.000	45.00
13	5.625	39.19
14	5.250	33.75
15	4.875	28.69
16	4.500	24.00
17	4.125	19.69
18	3.750	15.75
19	3.375	12.19
20	3.000	9.00
21	2.625	6.19
22	2.250	3.75
23	1.875	1.69
24	1.500	0

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TABLE II

CHARACTERIZATION OF 48 INCH LENGTH TANK PROBES FOR QUALIFICATION TESTS

Distance of fuel level below top of sensing portion (inches)	Dry capacitance per unit length (relative units)	Increase over dry capacitance (percent of total)
0	0.4	100.0
2	0.8	98.8
4	1.2	96.8
6	1.6	94.0
8	2.0	90.4
10	2.4	86.0
12	2.8	80.8
14	3.2	74.8
16	3.6	68.0
18	4.0	60.4
20	3.6	52.8
22	3.2	46.0
24	2.8	40.0
26	2.5	34.8
28	2.0	30.4
30	1.6	26.8
32	1.2	24.0
34	0.8	22.0
36	0.4	20.8
38	0.4	20.0
40	0.4-2.4	19.2
42	2.4	14.4
44	2.4	9.6
46	2.4	4.8
48	2.4	0

TABLE III

INDIVIDUAL TANK PROBE CALIBRATION DATA (See 6.2.1(d))

Height of fuel level in inches from bottom of tank probe	Added capacitance attributable to fuel, expressed in percent of total added capacitance for complete immersion		
	Tank Probe No. 1	Tank Probe No. 2	Additional tank probe as applicable

NOTE: Data for this table to be supplied by the contractor.

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4.6.7 Test Schedule - For acceptance under contract, the 1st and every 10th probe thereafter shall be tested in accordance with the Tank Probe Calibration Test except that Table III shall be used as the criterion. In the event a tank probe fails to meet the specified requirements, it shall be rejected. When a tank probe failure occurs, the next 10 consecutive tank probes manufactured shall be subjected to the Tank Probe Calibration Test.

4.6.8 Capacitance Stability Test - The capacitance of the tank probe or compensator probe shall be measured as received. The probe shall then be disassembled and reassembled ten times with capacitance measured after each reassembly. No one of the eleven measurements shall differ from any of the other by more than 0.2 percent. Due allowance for the error of the Standard bridge is to be considered so that it is not a part of the 0.2 percent.

4.6.9 Dielectric Strength Test - There shall be no breakdown of insulation of any of the equipment when it is subjected to a test voltage of 500V rms, 60 cps for a period of one minute applied between any pair of points desired by the testing activity.

4.6.10 Probe Capacitance at Low Temperature Test - The tank probe or compensator probe shall be placed in a temperature chamber and subjected to a temperature of -55° ± 2°C for a minimum period of 4 hours. At the end of this period, and while the probe is still at this temperature, the dry capacitance shall be determined and shall not differ from the respective "reference capacitance" by more than 1.0 percent.

4.6.10.1 When the probe has returned to room temperature, the dry capacitance shall again be determined and shall not differ from the respective "reference capacitance" by more than 0.5 percent.

*4.6.11 Capacitance at High Temperature Test - The tank probe or compensator probe shall be placed in a temperature chamber and subjected to a temperature of 125° ±3°C for a minimum period of 4 hours. At the end of this period, and while the probe is still at this temperature, the dry capacitance shall be determined and shall not differ from the respective "reference capacitance" by more than 1.0 percent and the electrical leakage of the probe shall meet the requirements of the Electrical Leakage Test.

4.6.12 Electrical Leakage Test - The electrical leakage resistance of the probe shall be determined, using an MD-2 or MD-2A tester. The resistance must be in excess of 500 megohms or numerically greater than $\frac{1}{\text{uuf}}$ x 50,000 expressed as megohms whichever is greater. In the terminology of the MD-2 or MD-2A tester, the values shall be taken between the following:

A - Ground . . . . . Hi-Z to ground

B - Ground . . . . . Lo-Z to ground

A - B . . . . . Hi-Z to Lo Z

4.6.12.1 When the probe has returned to room temperature, the dry capacitance shall again be determined and shall not differ from the respective "reference capacitance" by more than 0.5 percent.

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4.6.13 Vibration Test - The vibration test shall be conducted on the probes as specified below.

4.6.13.1 Probe Vibration Test - The tank probe or compensator probe shall be subjected to the frequency range of vibratory accelerations or double amplitudes not exceeding those shown below. (Vibration procedure XII of Specification MIL-E-5272, conducted at room temperature, may be substituted for this test).

Frequency in cpm	Double amplitude
300 to 600	0.050 inch
600 to 4,500	0.036 inch
4,500 to 30,000	10g

4.6.13.2 Probe Resonance Test - The tank probe or compensator probe shall be electrically connected to the master test instrument. The tank probe and the compensator probe shall be mounted as a separate item with the flange (for flange-mounted probes) fixed to the vibration stand, and with end supports and center supports substantially in accordance with types being provided in the respective airplane application and in accordance with the Reinforcement of Units requirements specified in Specification MIL-G-7940. Other types of probes shall be mounted on the test stand in a manner similar to that being followed in the respective airplane installation. The electrical and mechanical resonant frequencies of the probe(s) shall be determined while varying the frequency of applied variation slowly through the range of frequencies at vibratory accelerations not exceeding those specified. This procedure shall be followed successively for vibration applied along the 3 mutually perpendicular axes of the test specimen. The test specimen shall then be vibrated at the maximum electrical and mechanical resonant conditions for 2 hours (time specified refers to one axis of vibration) and with the applied double amplitude acceleration specified in the Probe Vibration Test. These periods of vibration shall be accomplished in sequence for each of the 3 mutually perpendicular axes of vibration. When more than one resonant frequency is encountered with vibration applied along any one axis, the test period shall be accomplished at the most severe resonance frequency. When resonant frequencies are not apparent within the specified frequency range, the specimen shall be vibrated twice as long as those specified for resonance at a frequency of 3,300 cpm and an applied double amplitude of 0.060 inch. The dry capacitance shall be determined after the test, and shall not differ from the "reference capacitance" by more than 0.5 percent. The probes shall be thoroughly inspected to determine that no damage has occurred as a result of the test.

4.6.14 Low Temperature Exposure Test - The tank probe or compensator probe shall be tested as described in the Probe Capacitance at Low Temperature Test except that the probe shall be maintained at a temperature of -65° ± 2°C for a period of 48 hours, and the temperature then raised to -54° ± 2°C for a period of 24 hours. At the completion of the 24-hour period, and while at -54°C, the dry capacitance shall be measured and shall not differ from the "reference capacitance" by more than 1.0 percent.

4.6.14.1 When the probe has returned to room temperature, the dry capacitance shall again be determined and shall not differ from the "reference capacitance" by more than 0.5 percent.

4.6.15 High Temperature Exposure Test - The tank probe or compensator probe shall be tested according to the capacitance at High Temperature Test except that the probe shall be maintained at the high temperature for a period of 24 hours. While still at this temperature, the capacitance shall be measured and shall not differ from the "reference capacitance" by more than 1.0 percent.

4.6.15.1 When the probe has returned to room temperature, the dry capacitance shall again be determined and shall not differ from the "reference capacitance" by more than 0.5 percent.

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4.6.16 Water Immersion - The tank probe or compensator probe shall be installed in a test tank as described below. The probe shall be mounted in its operating position, electrically connected to an MD-2 or MD-2A tester and resistive measurements made. The sensing portion of the tank probe shall then be covered to approximately three-quarters of its length with JP-5 fuel conforming to Specification MIL-J-5624. The compensator tank shall be filled to a level which will completely cover the compensator. Resistive and capacitive measurements shall be made and the fuel level noted. The MD-2 or MD-2A tester switch shall be set in the capacitive position. Sufficient distilled water shall be added to the tank to bring the fuel approximately midway between the top of the tank and the previous fuel level. The tank shall then be sealed, inverted for one minute and then restored to its starting position for 5 minutes to complete one cycle. The time required to invert the tank shall be 5 seconds. At the end of the fifth cycle, the water shall be drained slowly from the tank and, if necessary, fuel shall be added to restore the original fuel level. The capacitive reading shall then be noted. The change of indication from the original reading shall not exceed 2.0 percent of the original reading. Resistive readings shall be taken as described in the Electrical Leakage Test except that none of the readings shall be less than 100 megohms after the water immersion.

4.6.16.1 Tank Probe Test Tank - The probe shall be mounted in a suitable tank with a diameter of 5-1/2 ± 1/2 inches. The tank length shall be approximately 1-1/8 times the length of the probe. The tank shall be provided with fittings that will permit fuel and water to be added or drained without removing the probe.

4.6.16.2 Compensator Probe Test Tank - The tank shall be similar to the tank described in 4.6.16.1 except that its length shall be approximately 1-1/2 times the height of the compensator probe. The tank shall be made so that the probe can be installed at the bottom of the tank.

*4.6.17 Humidity Test - The tank probe or compensator probe shall be tested in accordance with Procedure I of Specification MIL-E-5272. One hour after completion of the exposure period, the dry capacitance shall be determined and shall not differ from the "reference capacitance" by more than 2.0 percent. Between 24 and 48 hours after the exposure period, the dry capacitance shall again be determined and shall not differ from the "reference capacitance" by more than 1.0 percent. The Electrical Leakage Test shall be repeated except that the probe shall be at room temperature. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the probes.

4.6.17.1 Within two hours of removal from humidity, the probe shall meet the requirements of the Electrical Leakage Test except that none of the readings shall be less than 100 megohms.

4.6.17.2 Between 24 and 48 hours after the exposure period, the probe shall meet the full requirements of the Electrical Leakage Test.

4.6.18 Fungus Resistance Test - The tank probe and compensator probe shall be tested in accordance with Procedure I of Specification MIL-E-5272 except that the exposure period shall be 14 days.

4.6.19 Salt Spray Test - All parts of the tank probe and compensator probe which are outside of the fuel cell shall be tested in accordance with Procedure I of Specification MIL-E-5272. One hour after completion of the exposure period, the dry capacitance shall be determined and shall not differ from the "referenced capacitance" by more than 2.0 percent. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the probes.

4.6.20 Fluid Resistance Test - The probe shall be soaked for 7 days in type II fuel of Specification MIL-S-3136 at a temperature of 70°C. At the end of this test, the probe shall be visually inspected to determine that no deterioration, damage, or defect has occurred caused by immersion in fuel. Following this test, the probe shall be subjected to the Water Immersion and Humidity tests.

4.6.21 Internal Examination Test - The probe shall be disassembled and all components shall be examined. Any deterioration or damage which could in any manner prevent the probe from meeting functional maintenance and service requirements during service life shall be cause for rejection.

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5. PREPARATION FOR DELIVERY

5.1 Packaging -

5.1.1 Level A - Each probe shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method IId, omitting preservation compound.

5.1.2 Level C - When this level is required, packaging shall be in accordance with standard commercial practice.

5.2 Packing -

5.2.1 Levels A and B - Each probe shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C - Each probe shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carrier's rules and regulations in effect at the time of shipment.

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary Marking - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended Use - The tank and compensator probes covered by this specification are intended for use in aircraft to sense the quantity of fuel contained in fuel tanks.

*6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number, and date of applicable specification.

(b) Type of probe (compensator or tank).

(c) Quantity and part number of probe.

(d) Whether Sampling Plan B is to be omitted and if not, add the name and address of the laboratory to which the samples should be submitted.

(e) Levels of packaging (see 5.1) and packing (see 5.2) desired.

6.2.1 Calibration Data for Acceptance Tests - The contractor should furnish the Inspector, on 8 by 10-1/2 inch paper, two copies of calibration data containing applicable information:

(a) Data for tank probe:

(1) Dry capacitance

(2) Total capacitance for complete immersion in nominal fuel

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(3) Added capacitance caused by complete immersion in nominal fuel.

(b) Data for compensator probe:

(1) Dry capacitance.

(2) Total capacitance for complete immersion in nominal fuel.

(3) Added capacitance caused by complete immersion in nominal fuel.

(c) Relative position of probes corresponding to locations in the airplane.

(d) Data to complete Table III.

6.3 Drawing Requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.4 Definitions -

6.4.1 Amplitude - The term "amplitude" as used herein is defined as the distance from the mean position to either extreme. "Double amplitude" is two times the "amplitude".

6.4.2 Nominal Fuel - Fuel with a dielectric constant of 2.041.

6.5 Provisions for Preproduction Tests - The manufacture of instruments on contract shall not commence until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of the probe from contract to contract, submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirement of acceptance testing.

6.5.1 It shall be understood that the instrument supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity. Evidence of unauthorized change shall constitute cause for rejection.

6.6 An asterisk preceding a paragraph denotes a change in that paragraph from the basic specification.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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